```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import zipfile
import os

# Path to your zip file
zip_path = "/content/drive/MyDrive/SMIT AI/Bone Fracture.zip"

# Folder where you want to extract the files
extract_to = "Bone Fracture yolo"

# Create the folder if it doesn't exist
os.makedirs(extract_to, exist_ok=True)

# Open and extract
with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall(extract_to)

print(f"Files extracted to: {extract_to}")
```

```
Files extracted to: Bone Fracture yolo
```

```
!pip install ultralytics
```

```
Collecting ultralytics
  Downloading ultralytics-8.3.187-py3-none-any.whl.metadata (37 kB)
Requirement already satisfied: numpy≥1.23.0 in /usr/local/lib/python3.12/dist-packages
(from ultralytics) (2.0.2)
Requirement already satisfied: matplotlib≥3.3.0 in /usr/local/lib/python3.12/dist-
packages (from ultralytics) (3.10.0)
Requirement already satisfied: opencv-python≥4.6.0 in /usr/local/lib/python3.12/dist-
packages (from ultralytics) (4.12.0.88)
Requirement already satisfied: pillow≥7.1.2 in /usr/local/lib/python3.12/dist-packages
(from ultralytics) (11.3.0)
Requirement already satisfied: pyyaml≥5.3.1 in /usr/local/lib/python3.12/dist-packages
(from ultralytics) (6.0.2)
Requirement already satisfied: requests≥2.23.0 in /usr/local/lib/python3.12/dist-
packages (from ultralytics) (2.32.4)
Requirement already satisfied: scipy≥1.4.1 in /usr/local/lib/python3.12/dist-packages
(from ultralytics) (1.16.1)
Requirement already satisfied: torch≥1.8.0 in /usr/local/lib/python3.12/dist-packages
(from ultralytics) (2.8.0+cu126)
Requirement already satisfied: torchvision≥0.9.0 in /usr/local/lib/python3.12/dist-
packages (from ultralytics) (0.23.0+cu126)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from
ultralytics) (5.9.5)
Requirement already satisfied: py-cpuinfo in /usr/local/lib/python3.12/dist-packages
(from ultralytics) (9.0.0)
Requirement already satisfied: polars in /usr/local/lib/python3.12/dist-packages (from
ultralytics) (1.25.2)
Collecting ultralytics-thop≥2.0.0 (from ultralytics)
  Downloading ultralytics_thop-2.0.16-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: contourpy≥1.0.1 in /usr/local/lib/python3.12/dist-
packages (from matplotlib≥3.3.0→ultralytics) (1.3.3)
Requirement already satisfied: cycler≥0.10 in /usr/local/lib/python3.12/dist-packages
(from matplotlib≥3.3.0→ultralytics) (0.12.1)
Requirement already satisfied: fonttools≥4.22.0 in /usr/local/lib/python3.12/dist-
packages (from matplotlib≥3.3.0→ultralytics) (4.59.1)
Requirement already satisfied: kiwisolver≥1.3.1 in /usr/local/lib/python3.12/dist-
packages (from matplotlib≥3.3.0→ultralytics) (1.4.9)
Requirement already satisfied: packaging≥20.0 in /usr/local/lib/python3.12/dist-
packages (from matplotlib≥3.3.0→ultralytics) (25.0)
Requirement already satisfied: pyparsing≥2.3.1 in /usr/local/lib/python3.12/dist-
packages (from matplotlib≥3.3.0→ultralytics) (3.2.3)
Requirement already satisfied: python-dateutil≥2.7 in /usr/local/lib/python3.12/dist-
packages (from matplotlib≥3.3.0→ultralytics) (2.9.0.post0)
Requirement already satisfied: charset_normalizer<4,≥2 in
/usr/local/lib/python3.12/dist-packages (from requests≥2.23.0→ultralytics) (3.4.3)
Requirement already satisfied: idna<4,≥2.5 in /usr/local/lib/python3.12/dist-packages
(from requests≥2.23.0→ultralytics) (3.10)
Requirement already satisfied: urllib3<3,≥1.21.1 in /usr/local/lib/python3.12/dist-
packages (from requests≥2.23.0→ultralytics) (2.5.0)
Requirement already satisfied: certifi≥2017.4.17 in /usr/local/lib/python3.12/dist-
packages (from requests≥2.23.0→ultralytics) (2025.8.3)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from
torch≥1.8.0→ultralytics) (3.19.1)
Requirement already satisfied: typing-extensions≥4.10.0 in
```

```
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (4.14.1)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages
(from torch≥1.8.0→ultralytics) (75.2.0)
Requirement already satisfied: sympy≥1.13.3 in /usr/local/lib/python3.12/dist-packages
(from torch≥1.8.0→ultralytics) (1.13.3)
Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-packages (from
torch≥1.8.0→ultralytics) (3.5)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from
torch≥1.8.0→ultralytics) (3.1.6)
Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from
torch≥1.8.0→ultralytics) (2025.3.0)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (12.6.77)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (12.6.77)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (12.6.80)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (12.6.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (11.3.0.4)
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (10.3.7.77)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (11.7.1.2)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (12.5.4.2)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.3 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (2.27.3)
Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (12.6.77)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (12.6.85)
Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in
/usr/local/lib/python3.12/dist-packages (from torch≥1.8.0→ultralytics) (1.11.1.6)
Requirement already satisfied: triton==3.4.0 in /usr/local/lib/python3.12/dist-packages
(from torch≥1.8.0→ultralytics) (3.4.0)
Requirement already satisfied: six≥1.5 in /usr/local/lib/python3.12/dist-packages (from
python-dateutil≥2.7→matplotlib≥3.3.0→ultralytics) (1.17.0)
Requirement already satisfied: mpmath<1.4,≥1.1.0 in /usr/local/lib/python3.12/dist-
packages (from sympy≥1.13.3→torch≥1.8.0→ultralytics) (1.3.0)
Requirement already satisfied: MarkupSafe≥2.0 in /usr/local/lib/python3.12/dist-
packages (from jinja2→torch≥1.8.0→ultralytics) (3.0.2)
Downloading ultralytics-8.3.187-py3-none-any.whl (1.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.1/1.1 MB 55.2 MB/s eta 0:00:00
Downloading ultralytics_thop-2.0.16-py3-none-any.whl (28 kB)
Installing collected packages: ultralytics-thop, ultralytics
Successfully installed ultralytics-8.3.187 ultralytics-thop-2.0.16
```

```
from ultralytics import YOLO
```

```
Creating new Ultralytics Settings v0.0.6 file ✅
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see
https://docs.ultralytics.com/quickstart/#ultralytics-settings.
```

```
model = YOLO("yolov8n.pt")
```

```
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt to
'yolov8n.pt': 100% ━━━━━━━━━━━━ 6.2/6.2MB 81.0MB/s 0.1s
```

```
yaml_path = "/content/Bone Fracture yolo/Human Bone Fractures Multi-modal Image Dataset
(HBFMID)/Bone Fractures Detection/data.yaml"
```

```
+model.train(
    data=yaml_path,
    epochs=100,
    imgsz=640,
    batch=16,
    patience=20,
    pretrained = True
)
```

```
Ultralytics 8.3.187 🚀 Python-3.12.11 torch-2.8.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugment,
batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10,
cls=0.5, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0,
data=/content/Bone Fracture yolo/Human Bone Fractures Multi-modal Image Dataset
(HBFMID)/Bone Fractures Detection/data.yaml, degrees=0.0, deterministic=True,
device=None, dfl=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=100,
erasing=0.4, exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0,
freeze=None, half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False,
iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4,
max_det=300, mixup=0.0, mode=train, model=yolov8n.pt, momentum=0.937, mosaic=1.0,
multi_scale=False, name=train, nbs=64, nms=False, opset=None, optimize=False,
optimizer=auto, overlap_mask=True, patience=20, perspective=0.0, plots=True, pose=12.0,
pretrained=True, profile=False, project=None, rect=False, resume=False,
retina_masks=False, save=True, save_conf=False, save_crop=False,
save_dir=runs/detect/train, save_frames=False, save_json=False, save_period=-1,
save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True,
show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None,
split=val, stream_buffer=False, task=detect, time=None, tracker=botsort.yaml,
translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False,
warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005,
workers=8, workspace=None
Downloading https://ultralytics.com/assets/Arial.ttf to
'/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━ 755.1/755.1KB 13.7MB/s 0.1s
Overriding model.yaml nc=80 with nc=10

                   from  n    params  module
arguments
  0                  -1  1       464  ultralytics.nn.modules.conv.Conv             [3,
16, 3, 2]
  1                  -1  1      4672  ultralytics.nn.modules.conv.Conv             [16,
32, 3, 2]
  2                  -1  1      7360  ultralytics.nn.modules.block.C2f             [32,
32, 1, True]
  3                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32,
64, 3, 2]
  4                  -1  2     49664  ultralytics.nn.modules.block.C2f             [64,
64, 2, True]
  5                  -1  1     73984  ultralytics.nn.modules.conv.Conv             [64,
128, 3, 2]
  6                  -1  2    197632  ultralytics.nn.modules.block.C2f             [128,
128, 2, True]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128,
256, 3, 2]
  8                  -1  1    460288  ultralytics.nn.modules.block.C2f             [256,
256, 1, True]
```

```
  9                  -1  1    164608  ultralytics.nn.modules.block.SPPF            [256,
256, 5]
 10                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 11             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 12                  -1  1    148224  ultralytics.nn.modules.block.C2f             [384,
128, 1]
 13                  -1  1         0  torch.nn.modules.upsampling.Upsample
[None, 2, 'nearest']
 14             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 15                  -1  1     37248  ultralytics.nn.modules.block.C2f             [192,
64, 1]
 16                  -1  1     36992  ultralytics.nn.modules.conv.Conv             [64,
64, 3, 2]
 17            [-1, 12]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 18                  -1  1    123648  ultralytics.nn.modules.block.C2f             [192,
128, 1]
 19                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128,
128, 3, 2]
 20             [-1, 9]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 21                  -1  1    493056  ultralytics.nn.modules.block.C2f             [384,
256, 1]
 22        [15, 18, 21]  1    753262  ultralytics.nn.modules.head.Detect           [10,
[64, 128, 256]]
Model summary: 129 layers, 3,012,798 parameters, 3,012,782 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt to
'yolo11n.pt': 100% ━━━━━━━━━━━━ 5.4/5.4MB 104.1MB/s 0.1s
AMP: checks passed ✅
train: Fast image access ✅ (ping: 0.0±0.0 ms, read: 570.0±204.0 MB/s, size: 26.7 KB)
train: Scanning /content/Bone Fracture yolo/Human Bone Fractures Multi-modal Image
Dataset (HBFMID)/Bone Fractures Detection/train/labels... 1347 images, 3 backgrounds, 0
corrupt: 100% ━━━━━━━━━━━━ 1347/1347 1741.8it/s 0.8s
train: New cache created: /content/Bone Fracture yolo/Human Bone Fractures Multi-modal
Image Dataset (HBFMID)/Bone Fractures Detection/train/labels.cache
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)),
ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01,
clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))
val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 351.8±220.4 MB/s, size: 25.1 KB)
val: Scanning /content/Bone Fracture yolo/Human Bone Fractures Multi-modal Image Dataset
(HBFMID)/Bone Fractures Detection/valid/labels... 128 images, 0 backgrounds, 0 corrupt:
100% ━━━━━━━━━━━━ 128/128 1809.6it/s 0.1s
val: New cache created: /content/Bone Fracture yolo/Human Bone Fractures Multi-modal
Image Dataset (HBFMID)/Bone Fractures Detection/valid/labels.cache
Plotting labels to runs/detect/train/labels.jpg...
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and
determining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.000714, momentum=0.9) with parameter groups 57 weight(decay=0.0),
64 weight(decay=0.0005), 63 bias(decay=0.0)
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to runs/detect/train
Starting training for 100 epochs...
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      1/100      2.13G      2.561      5.412      2.404          5        640: 100%
━━━━━━━━━━━━ 85/85 3.1it/s 27.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.1it/s 1.9s
                   all        128        157       0.68     0.0803     0.0711     0.0312

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      2/100       2.6G      2.143      4.293       1.99          9        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.0it/s 2.0s
                   all        128        157      0.776     0.0766     0.0977     0.0399

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      3/100      2.63G      2.075       3.87      1.953          3        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.3it/s 1.7s
                   all        128        157      0.866     0.0805      0.149     0.0747

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      4/100      2.63G       2.02      3.488      1.919          5        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.5s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.0it/s 1.3s
                   all        128        157       0.59      0.242      0.254      0.117

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      5/100      2.66G       2.01      3.139      1.914          5        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.8it/s 1.4s
                   all        128        157      0.645       0.15       0.17     0.0725

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      6/100      2.68G      1.962      2.888      1.897          5        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.5s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.5it/s 1.1s
                   all        128        157      0.632      0.259      0.303      0.149

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      7/100       2.7G      1.933      2.692      1.829          4        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.3it/s 1.8s
                   all        128        157      0.706      0.183      0.242      0.126

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      8/100       2.7G      1.912      2.544       1.85          5        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.1it/s 1.3s
                   all        128        157      0.461      0.316       0.31       0.15
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      9/100      2.73G      1.836      2.351      1.763          3        640: 100%
━━━━━━━━━━━━ 85/85 3.4it/s 24.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.5it/s 1.2s
                   all        128        157      0.385      0.427      0.369      0.175

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     10/100      2.75G      1.822      2.318      1.767          5        640: 100%
━━━━━━━━━━━━ 85/85 3.4it/s 24.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.2it/s 1.2s
                   all        128        157      0.427      0.329      0.416      0.186

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     11/100      2.76G      1.794      2.144      1.731          5        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 23.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.1it/s 1.9s
                   all        128        157      0.582       0.39        0.4      0.193

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     12/100      2.77G      1.774      2.006      1.706          4        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 23.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.1it/s 1.3s
                   all        128        157      0.628       0.32      0.466      0.218

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     13/100       2.8G      1.767      1.937      1.694          4        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.2it/s 1.2s
                   all        128        157      0.602        0.4      0.504      0.204

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     14/100      2.81G      1.734      1.812      1.649          9        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.2it/s 1.3s
                   all        128        157      0.448      0.404      0.509      0.253

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     15/100      2.83G      1.754      1.841      1.661          5        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.1s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.7it/s 1.5s
                   all        128        157      0.707       0.46      0.556      0.274

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     16/100      2.84G      1.687       1.75      1.619          5        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.3it/s 1.7s
                   all        128        157      0.687      0.523      0.607      0.277
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     17/100      2.87G      1.702      1.729      1.641          3        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.3it/s 1.2s
                   all        128        157      0.732      0.558      0.622      0.331

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     18/100      2.88G      1.673      1.665      1.591          4        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.2it/s 1.3s
                   all        128        157      0.671       0.52       0.66      0.346

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     19/100       2.9G       1.66      1.614      1.606          8        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.1s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.3it/s 1.2s
                   all        128        157      0.738      0.535       0.64       0.29

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     20/100      2.91G       1.61      1.567      1.572          6        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.1it/s 1.9s
                   all        128        157      0.785      0.552      0.741      0.353

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     21/100      2.93G      1.593      1.505      1.539          7        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 23.1s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.2it/s 1.2s
                   all        128        157      0.655      0.684      0.722      0.344

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     22/100      2.95G      1.602      1.484       1.56          4        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.5it/s 1.1s
                   all        128        157      0.517      0.656      0.663      0.315

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     23/100      2.97G      1.571      1.435      1.541          5        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.5it/s 1.1s
                   all        128        157      0.748        0.4      0.527      0.271

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     24/100      2.97G      1.559      1.405      1.517          9        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 1.9it/s 2.1s
                   all        128        157       0.83      0.669      0.813      0.406
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     25/100         3G       1.53      1.334      1.499          4        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.0it/s 1.3s
                   all        128        157      0.785      0.604      0.783      0.393

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     26/100      3.02G      1.535      1.307      1.504          4        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.5s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.0it/s 1.3s
                   all        128        157      0.786      0.623      0.753      0.383

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     27/100      3.04G      1.513       1.31      1.481          7        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.2it/s 1.2s
                   all        128        157      0.708      0.711      0.776      0.388

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     28/100      3.04G      1.515      1.307      1.491          4        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.1it/s 1.3s
                   all        128        157      0.741      0.769      0.812      0.445

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     29/100      3.07G      1.544      1.269      1.488         12        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.2it/s 1.8s
                   all        128        157      0.552      0.816      0.807      0.411

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     30/100      3.09G      1.501      1.224      1.465          3        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 23.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.2it/s 1.3s
                   all        128        157      0.819      0.577      0.786      0.403

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     31/100       3.1G      1.466       1.18       1.43          5        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.5s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.8it/s 1.1s
                   all        128        157      0.791      0.732      0.868      0.438

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     32/100      3.11G      1.465      1.178      1.437          8        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.3it/s 1.2s
                   all        128        157      0.857      0.701      0.861       0.48
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     33/100      3.14G      1.475      1.173       1.46          7        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 1.8it/s 2.2s
                   all        128        157      0.787      0.718      0.836      0.429

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     34/100      3.16G       1.46      1.151      1.433          8        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.2it/s 1.3s
                   all        128        157      0.811      0.735      0.867      0.441

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     35/100      3.17G      1.452      1.144      1.421          7        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.5it/s 1.1s
                   all        128        157       0.82      0.759      0.863      0.461

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     36/100      3.18G      1.439      1.106      1.413          6        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.3it/s 1.2s
                   all        128        157      0.819      0.775      0.863      0.463

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     37/100      3.21G      1.391      1.083      1.378          5        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.5it/s 1.6s
                   all        128        157      0.844      0.827      0.872      0.509

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     38/100      3.22G      1.409      1.047      1.392          5        640: 100%
━━━━━━━━━━━━ 85/85 3.8it/s 22.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.3it/s 1.7s
                   all        128        157      0.816      0.835      0.879      0.449

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     39/100      3.24G      1.406      1.057      1.386          3        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.1it/s 1.3s
                   all        128        157      0.925      0.654      0.844      0.448

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     40/100      3.25G      1.419       1.06      1.413          2        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.9it/s 1.4s
                   all        128        157      0.846      0.768      0.873      0.484
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     41/100      3.28G      1.385      1.025      1.386          7        640: 100%
━━━━━━━━━━━━ 85/85 3.4it/s 25.2s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.5it/s 1.1s
                   all        128        157      0.804       0.81      0.881       0.48

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     42/100      3.29G      1.361     0.9908      1.355          8        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.0s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 1.9it/s 2.1s
                   all        128        157      0.949      0.761      0.888      0.492

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     43/100      3.31G      1.375       1.01      1.365          4        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.6it/s 1.5s
                   all        128        157      0.826      0.799      0.864      0.475

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     44/100      3.32G      1.366       1.03       1.36          6        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.2it/s 1.2s
                   all        128        157      0.898      0.831      0.902       0.48

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     45/100      3.34G      1.325      1.001      1.347          9        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.3s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.3it/s 1.2s
                   all        128        157      0.789      0.849      0.885      0.472

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     46/100      3.36G      1.339      0.965      1.346          4        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.6it/s 1.1s
                   all        128        157      0.714      0.843      0.874      0.475

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     47/100      3.38G      1.334     0.9692      1.338          4        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 1.9it/s 2.1s
                   all        128        157      0.903      0.837      0.902      0.505

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     48/100      3.38G      1.342     0.9867      1.344          9        640: 100%
━━━━━━━━━━━━ 85/85 3.7it/s 22.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.5it/s 1.1s
                   all        128        157      0.918      0.833      0.897      0.493
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     49/100      3.41G      1.327     0.9437      1.345          2        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.4it/s 1.2s
                   all        128        157      0.895      0.841      0.896      0.494

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     50/100      3.43G      1.297     0.9096       1.31          5        640: 100%
━━━━━━━━━━━━ 85/85 3.4it/s 24.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.8it/s 1.0s
                   all        128        157      0.906      0.859        0.9      0.498

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     51/100      3.45G      1.267     0.8836      1.288          8        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.0it/s 1.3s
                   all        128        157      0.871      0.824      0.892      0.481

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     52/100      3.45G      1.282     0.8962       1.31          7        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.7s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.0it/s 2.0s
                   all        128        157      0.867      0.814      0.895      0.465

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     53/100      3.48G      1.281     0.8949      1.309          8        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.6s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.5it/s 1.1s
                   all        128        157      0.871      0.857      0.887      0.451

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     54/100       3.5G      1.301     0.9111      1.315          3        640: 100%
━━━━━━━━━━━━ 85/85 3.5it/s 24.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.3it/s 1.2s
                   all        128        157      0.757      0.852      0.827      0.459

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     55/100      3.52G      1.248      0.867      1.282          6        640: 100%
━━━━━━━━━━━━ 85/85 3.4it/s 24.9s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 3.5it/s 1.2s
                   all        128        157      0.763      0.847      0.899      0.495

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     56/100      3.52G      1.268     0.8595      1.304          5        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.8s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.2it/s 1.8s
                   all        128        157      0.891      0.808      0.895      0.481
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
     57/100      3.55G      1.245     0.8689      1.285          4        640: 100%
━━━━━━━━━━━━ 85/85 3.6it/s 23.4s
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 2.1it/s 1.9s
                   all        128        157      0.869      0.851      0.915      0.494
EarlyStopping: Training stopped early as no improvement observed in last 20 epochs. Best
results observed at epoch 37, best model saved as best.pt.
To update EarlyStopping(patience=20) pass a new patience value, i.e. `patience=300` or
use `patience=0` to disable EarlyStopping.

57 epochs completed in 0.407 hours.
Optimizer stripped from runs/detect/train/weights/last.pt, 6.3MB
Optimizer stripped from runs/detect/train/weights/best.pt, 6.3MB

Validating runs/detect/train/weights/best.pt...
Ultralytics 8.3.187 🚀 Python-3.12.11 torch-2.8.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 72 layers, 3,007,598 parameters, 0 gradients, 8.1 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-
95): 100% ━━━━━━━━━━━━ 4/4 1.4it/s 2.9s
                   all        128        157      0.844      0.828      0.869       0.51
            Comminuted         15         15      0.947      0.667      0.748      0.454
            Greenstick         10         10          1      0.397       0.45      0.193
               Healthy          7          7      0.842          1      0.995      0.727
                Linear          1          1      0.697          1      0.995      0.597
    Oblique Displaced         30         30      0.869      0.666       0.85      0.417
               Oblique          7          7      0.747      0.857      0.913      0.544
             Segmental          3          3      0.986          1      0.995      0.863
                Spiral          5          5      0.683          1      0.962      0.486
  Transverse Displaced         53         65      0.844      0.835      0.868      0.358
            Transverse         11         14      0.821      0.857      0.914      0.457
Speed: 0.4ms preprocess, 2.7ms inference, 0.0ms loss, 7.2ms postprocess per image
Results saved to runs/detect/train
```

```
ultralytics.utils.metrics.DetMetrics object with attributes:

ap_class_index: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at 0x7cf37e143980>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)', 'Recall-
Confidence(B)']
curves_results: [[array([          0,    0.001001,    0.002002,    0.003003,
0.004004,    0.005005,    0.006006,    0.007007,    0.008008,    0.009009,     0.01001,
0.011011,    0.012012,    0.013013,    0.014014,    0.015015,    0.016016,    0.017017,
0.018018,    0.019019,     0.02002,    0.021021,    0.022022,    0.023023,
          0.024024,    0.025025,    0.026026,    0.027027,    0.028028,    0.029029,
0.03003,    0.031031,    0.032032,    0.033033,    0.034034,    0.035035,    0.036036,
0.037037,    0.038038,    0.039039,     0.04004,    0.041041,    0.042042,    0.043043,
0.044044,    0.045045,    0.046046,    0.047047,
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0.054054,    0.055055,    0.056056,    0.057057,    0.058058,    0.059059,     0.06006,
0.061061,    0.062062,    0.063063,    0.064064,    0.065065,    0.066066,    0.067067,
0.068068,    0.069069,     0.07007,    0.071071,
          0.072072,    0.073073,    0.074074,    0.075075,    0.076076,    0.077077,
0.078078,    0.079079,     0.08008,    0.081081,    0.082082,    0.083083,    0.084084,
```

```
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          0.096096,    0.097097,    0.098098,    0.099099,      0.1001,      0.1011,
0.1021,      0.1031,      0.1041,     0.10511,     0.10611,     0.10711,     0.10811,
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0.017017,    0.018018,    0.019019,     0.02002,    0.021021,    0.022022,    0.023023,
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0.998,       0.999,           1]), array([[        0.8,         0.8,     0.73333, ...,
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0],
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0],
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0],
       [          1,           1,           1, ...,           0,           0,
0]]), 'Confidence', 'Recall']]
fitness: np.float64(0.5455510268705388)
keys: ['metrics/precision(B)', 'metrics/recall(B)', 'metrics/mAP50(B)', 'metrics/mAP50-
95(B)']
maps: array([    0.45385,     0.19279,     0.72713,       0.597,     0.41732,
0.54364,     0.86305,     0.48627,     0.35786,     0.45726])
names: {0: 'Comminuted', 1: 'Greenstick', 2: 'Healthy', 3: 'Linear', 4: 'Oblique
Displaced', 5: 'Oblique', 6: 'Segmental', 7: 'Spiral', 8: 'Transverse Displaced', 9:
'Transverse'}
nt_per_class: array([15, 10,  7,  1, 30,  7,  3,  5, 65, 14])
nt_per_image: array([15, 10,  7,  1, 30,  7,  3,  5, 53, 11])
results_dict: {'metrics/precision(B)': 0.8437850461987981, 'metrics/recall(B)':
0.8278231606114677, 'metrics/mAP50(B)': 0.8689507138933182, 'metrics/mAP50-95(B)':
0.5096177283124521, 'fitness': 0.5455510268705388}
save_dir: PosixPath('runs/detect/train')
speed: {'preprocess': 0.3920365234382217, 'inference': 2.707421476562999, 'loss':
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stats: {'tp': [], 'conf': [], 'pred_cls': [], 'target_cls': [], 'target_img': []}
task: 'detect'
```